



11995 - Exploring the surface evolution of Centaurs: An empirical test of solar system dynamical history

Cycle: 5, Proposal Category: GO

INVESTIGATORS

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OBSERVATIONS

<i>Folder</i>	<i>Observation</i>	<i>Label</i>	<i>Observing Template</i>	<i>Science Target</i>
Observation Folder				
	1	2016LN8	NIRSpec IFU Spectroscopy	(1) 2016LN8
	2	2000EC98	NIRSpec IFU Spectroscopy	(2) 2000EC98
	3	2001BL41	NIRSpec IFU Spectroscopy	(3) 2001BL41
	4	2023JF4	NIRSpec IFU Spectroscopy	(4) 2023JF4
	5	2023QB47	NIRSpec IFU Spectroscopy	(5) 2023QB47
	6	2020RR5	NIRSpec IFU Spectroscopy	(6) 2020RR5
	7	2021DB17	NIRSpec IFU Spectroscopy	(7) 2021DB17

<i>Folder</i>	<i>Observation</i>	<i>Label</i>	<i>Observing Template</i>	<i>Science Target</i>
	8	2014UJ225	NIRSpec IFU Spectroscopy	(8) 2014UJ225

ABSTRACT

Current theories of solar system evolution posit that the Kuiper belt objects (KBOs) and Jupiter Trojans share a common outer solar system origin. However, recent JWST observations have presented a conundrum: while KBO spectra show a rich array of spectroscopic features, Trojan spectra are mostly featureless. In order to reconcile this discrepancy within our current understanding of solar system history, some process must have altered the Trojans' surfaces during their inward migration. Centaurs - inward-scattered KBOs on orbits that cross the giant planet region - are the ideal proxy population for addressing this question. JWST spectra of Centaurs hint at a progression in spectral shape that bridges the gap between the surface properties of KBOs and Trojans. Specifically, a handful of low-perihelion Centaurs show starkly attenuated features when compared to the more distant Centaurs and KBOs, suggesting that thermophysical processing of the surface may be the key to explaining the Trojans' muted spectral features. However, the current sample of close-in Centaurs is limited by small number statistics and size-dependent selection biases. To properly investigate this emergent trend, we propose to obtain NIRSpec PRISM mode spectra of 8 additional Centaurs. By confirming the signal of thermal evolution in the spectral features of close-in Centaurs, we will obtain a definitive explanation for the discrepant KBO and Trojan spectra, which would serve as a powerful empirical validation of current dynamical instability models of solar system evolution.

OBSERVING DESCRIPTION

We will observe 8 inactive Centaurs with the NIRSpec IFU. The targets include 5 objects with $q < 9$ au and 3 small high-perihelion Centaurs with $q > 12$ au and $H > 10$. The goal of these observations will be to search for distinctive attenuated spectral features at 3 and 4.25 microns among the low-perihelion Centaurs.

The spectra will be obtained using the low-resolution PRISM grating, which will provide continuous coverage from 0.6 to 5.3 microns. We require $\text{SNR} = 15$ at 4.25 microns to detect carbon dioxide features on the low-perihelion Centaurs with depths as low as 10%. For the fainter high-perihelion targets, the observations are designed to provide $\text{SNR} = 15$ at 3.0 microns, which is sufficient to reliably measure 3-micron band depths as low as 10%. Our observing strategy follows standard procedures for faint moving target observations. We will collect a set dithered single-integration exposures, utilizing the NRSIRS2RAPID readout and a 4-point dither pattern to minimize readout noise and detector effects.

Proposal 11995 - Targets - Exploring the surface evolution of Centaurs: An empirical test of solar system dynamical history

Solar System Targets	#	Name	Level 1	Level 2	Level 3
	(1)	2016LN8	TYPE=ASTEROID,A=5.756387706771593,E=0.0661 3905668755486,I=43.34223452516974,O=87.3981337 3223463,W=16.5391729568454,M=341.59707307453 81,EQUINOX=J2000,EPOCH=21-NOV- 2025:00:00:00,EpochTimeScale=TDB		
		Comments: Low-q Centaur Extended=NO			
	(2)	2000EC98	TYPE=ASTEROID,A=10.7587993793109,E=0.45445 71494047505,I=4.340035613712524,O=173.26318167 2681,W=163.4952961846111,M=108.0953867884616, EQUINOX=J2000,EPOCH=21-NOV- 2025:00:00:00,EpochTimeScale=TDB		
		Comments: Low-q Centaur Extended=NO			
	(3)	2001BL41	TYPE=ASTEROID,A=9.873184918953188,E=0.2959 575105306945,I=12.40246682166828,O=280.4228535 494468,W=129.7769895610884,M=332.70847893503 26,EQUINOX=J2000,EPOCH=21-NOV- 2025:00:00:00,EpochTimeScale=TDB		
		Comments: Low-q Centaur Extended=NO			
	(4)	2023JF4	TYPE=ASTEROID,A=12.77596921801687,E=0.4092 273638915986,I=15.64781116342964,O=142.2267256 634491,W=202.9923131629253,M=323.43113005211 64,EQUINOX=J2000,EPOCH=21-NOV- 2025:00:00:00,EpochTimeScale=TDB		
		Comments: Low-q Centaur Extended=NO			
	(5)	2023QB47	TYPE=ASTEROID,A=18.94384750780507,E=0.6653 745839733894,I=21.20608724281536,O=331.2153559 072036,W=78.46534405478047,M=359.36593345493 81,EQUINOX=J2000,EPOCH=21-NOV- 2025:00:00:00,EpochTimeScale=TDB		
		Comments: Low-q Centaur Extended=NO			
	(6)	2020RR5	TYPE=ASTEROID,A=22.68024715492194,E=0.4612 797396294905,I=13.2387197034441,O=133.45585933 79135,W=261.5714748223553,M=7.43105470006362 6,EQUINOX=J2000,EPOCH=21-NOV- 2025:00:00:00,EpochTimeScale=TDB		
		Comments: Small Centaur with q>12 AU Extended=NO			
	(7)	2021DB17	TYPE=ASTEROID,A=19.15690537839405,E=0.3581 399317267978,I=7.508620307976723,O=28.09335608 108688,W=177.3037502569899,M=8.5362258821900 33,EQUINOX=J2000,EPOCH=21-NOV- 2025:00:00:00,EpochTimeScale=TDB		
		Comments: Small Centaur with q>12 AU Extended=NO			
	(8)	2014UJ225	TYPE=ASTEROID,A=23.42363096764399,E=0.3836 924131393097,I=21.30793360463058,O=18.89883633 455016,W=82.19947636626529,M=1.0498142035173 41,EQUINOX=J2000,EPOCH=21-NOV- 2025:00:00:00,EpochTimeScale=TDB		
		Comments: Small Centaur with q>12 AU Extended=NO			

Proposal 11995 - Observation 1 - Exploring the surface evolution of Centaurs: An empirical test of solar system dynamical history

Observation	Proposal 11995, Observation 1: 2016LN8											Thu Jun 04 15:00:12 GMT 2026
	Diagnostic Status: Warning											
	Observing Template: NIRSpec IFU Spectroscopy											
Diagnostics	(Visit 1:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.											
	(2016LN8 (Obs 1)) Informational (Form): The Visit Planner and Spike may produce different schedulability results.											
Solar System Targets	#	Name	Level 1				Level 2				Level 3	
	(1)	2016LN8	TYPE=ASTEROID,A=5.756387706771593,E=0.0661 3905668755486,I=43.34223452516974,O=87.3981337 3223463,W=16.5391729568454,M=341.59707307453 81,EQUINOX=J2000,EPOCH=21-NOV- 2025:00:00:00,EpochTimeScale=TDB									
	Comments: Low-q Centaur Extended=NO											
Template	TA Method						HFF Readout Mode					
	NONE						false					
Dithers	#	Dither Type		Size		Starting Point		Number of Points		Points		
	1	4-POINT-DITHER										
Spectral Elements	#	Grating/Filter	Readout Pattern	Groups/Int	Integrations/Exp	Leakcal	Dither	Autocal	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
	1	PRISM/CLEAR	NRSIRS2RAPID	28	1	false	true	NONE	4	4	1692.311	
Special Requirements	DEFAULT WINDOW: ANGULAR RATE 2016LN8 FROM JWST LESS THAN 0.075											

Proposal 11995 - Observation 2 - Exploring the surface evolution of Centaurs: An empirical test of solar system dynamical history

Observation	Proposal 11995, Observation 2: 2000EC98											Thu Jun 04 15:00:12 GMT 2026
	Diagnostic Status: Warning											
Observing Template: NIRSpec IFU Spectroscopy												
Diagnostics	(Visit 2:1) Warning (Form): Overheads are provisional until the Visit Planner has been run. (2000EC98 (Obs 2)) Informational (Form): The Visit Planner and Spike may produce different schedulability results.											
Solar System Targets	#	Name	Level 1				Level 2				Level 3	
	(2)	2000EC98	TYPE=ASTEROID,A=10.7587993793109,E=0.45445 71494047505,I=4.340035613712524,O=173.26318167 2681,W=163.4952961846111,M=108.0953867884616, EQUINOX=J2000,EPOCH=21-NOV- 2025:00:00:00,EpochTimeScale=TDB <i>Comments: Low-q Centaur Extended=NO</i>									
Template	TA Method						HFF Readout Mode					
	NONE						false					
Dithers	#	Dither Type		Size		Starting Point		Number of Points		Points		
	1	4-POINT-DITHER										
Spectral Elements	#	Grating/Filter	Readout Pattern	Groups/Int	Integrations/Exp	Leakcal	Dither	Autocal	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
	1	PRISM/CLEAR	NRSIRS2RAPID	37	1	false	true	NONE	4	4	2217.511	
Special Requirements	DEFAULT WINDOW: ANGULAR RATE 2000EC98 FROM JWST LESS THAN 0.075											

Proposal 11995 - Observation 3 - Exploring the surface evolution of Centaurs: An empirical test of solar system dynamical history

Observation	Proposal 11995, Observation 3: 2001BL41											Thu Jun 04 15:00:12 GMT 2026
	Diagnostic Status: Warning											
	Observing Template: NIRSpec IFU Spectroscopy											
Diagnostics	(Visit 3:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.											
	(2001BL41 (Obs 3)) Informational (Form): The Visit Planner and Spike may produce different schedulability results.											
Solar System Targets	#	Name	Level 1				Level 2				Level 3	
	(3)	2001BL41	TYPE=ASTEROID,A=9.873184918953188,E=0.2959 575105306945,I=12.40246682166828,O=280.4228535 494468,W=129.7769895610884,M=332.70847893503 26,EQUINOX=J2000,EPOCH=21-NOV- 2025:00:00:00,EpochTimeScale=TDB									
	Comments: Low-q Centaur Extended=NO											
Template	TA Method						HFF Readout Mode					
	NONE						false					
Dithers	#	Dither Type			Size	Starting Point			Number of Points		Points	
	1	4-POINT-DITHER										
Spectral Elements	#	Grating/Filter	Readout Pattern	Groups/Int	Integrations/Exp	Leakcal	Dither	Autocal	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
	1	PRISM/CLEAR	NRSIRS2RAPID	24	1	false	true	NONE	4	4	1458.889	
Special Requirements	DEFAULT WINDOW: ANGULAR RATE 2001BL41 FROM JWST LESS THAN 0.075											

Proposal 11995 - Observation 4 - Exploring the surface evolution of Centaurs: An empirical test of solar system dynamical history

Observation	Proposal 11995, Observation 4: 2023JF4											Thu Jun 04 15:00:12 GMT 2026
	Diagnostic Status: Warning											
	Observing Template: NIRSpec IFU Spectroscopy											
Diagnostics	(Visit 4:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.											
	(2023JF4 (Obs 4)) Informational (Form): The Visit Planner and Spike may produce different schedulability results.											
Solar System Targets	#	Name	Level 1				Level 2				Level 3	
	(4)	2023JF4	TYPE=ASTEROID,A=12.77596921801687,E=0.4092 273638915986,I=15.64781116342964,O=142.2267256 634491,W=202.9923131629253,M=323.43113005211 64,EQUINOX=J2000,EPOCH=21-NOV- 2025:00:00:00,EpochTimeScale=TDB									
	Comments: Low-q Centaur Extended=NO											
Template	TA Method						HFF Readout Mode					
	NONE						false					
Dithers	#	Dither Type			Size	Starting Point			Number of Points		Points	
	1	4-POINT-DITHER										
Spectral Elements	#	Grating/Filter	Readout Pattern	Groups/Int	Integrations/Exp	Leakcal	Dither	Autocal	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
	1	PRISM/CLEAR	NRSIRS2RAPID	41	1	false	true	NONE	4	4	2450.934	
Special Requirements	DEFAULT WINDOW: ANGULAR RATE 2023JF4 FROM JWST LESS THAN 0.075											

Proposal 11995 - Observation 5 - Exploring the surface evolution of Centaurs: An empirical test of solar system dynamical history

Observation	Proposal 11995, Observation 5: 2023QB47											Thu Jun 04 15:00:12 GMT 2026
	Diagnostic Status: Warning											
Observing Template: NIRSpec IFU Spectroscopy												
Diagnostics	(Visit 5:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.											
	(2023QB47 (Obs 5)) Informational (Form): The Visit Planner and Spike may produce different schedulability results.											
Solar System Targets	#	Name	Level 1				Level 2				Level 3	
	(5)	2023QB47	TYPE=ASTEROID,A=18.94384750780507,E=0.6653745839733894,I=21.20608724281536,O=331.2153559072036,W=78.46534405478047,M=359.3659334549381,EQUINOX=J2000,EPOCH=21-NOV-2025:00:00:00,EpochTimeScale=TDB									
Comments: Low-q Centaur Extended=NO												
Template	TA Method						HFF Readout Mode					
	NONE						false					
Dithers	#	Dither Type		Size		Starting Point		Number of Points		Points		
	1	4-POINT-DITHER										
Spectral Elements	#	Grating/Filter	Readout Pattern	Groups/Int	Integrations/Exp	Leakcal	Dither	Autocal	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
	1	PRISM/CLEAR	NRSIRS2RAPID	81	1	false	true	NONE	4	4	4785.156	
Special Requirements	DEFAULT WINDOW: ANGULAR RATE 2023QB47 FROM JWST LESS THAN 0.075											

Proposal 11995 - Observation 6 - Exploring the surface evolution of Centaurs: An empirical test of solar system dynamical history

Observation	Proposal 11995, Observation 6: 2020RR5											Thu Jun 04 15:00:12 GMT 2026
	Diagnostic Status: Warning											
	Observing Template: NIRSpec IFU Spectroscopy											
Diagnostics	(Visit 6:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.											
	(2020RR5 (Obs 6)) Informational (Form): The Visit Planner and Spike may produce different schedulability results.											
Solar System Targets	#	Name	Level 1				Level 2				Level 3	
	(6)	2020RR5	TYPE=ASTEROID,A=22.68024715492194,E=0.4612 797396294905,I=13.2387197034441,O=133.45585933 79135,W=261.5714748223553,M=7.43105470006362 6,EQUINOX=J2000,EPOCH=21-NOV- 2025:00:00:00,EpochTimeScale=TDB									
	Comments: Small Centaur with q>12 AU Extended=NO											
Template	TA Method						HFF Readout Mode					
	NONE						false					
Dithers	#	Dither Type			Size	Starting Point			Number of Points		Points	
	1	4-POINT-DITHER										
Spectral Elements	#	Grating/Filter	Readout Pattern	Groups/Int	Integrations/Exp	Leakcal	Dither	Autocal	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
	1	PRISM/CLEAR	NRSIRS2RAPID	38	1	false	true	NONE	4	4	2275.867	
Special Requirements	DEFAULT WINDOW: ANGULAR RATE 2020RR5 FROM JWST LESS THAN 0.075											

Proposal 11995 - Observation 7 - Exploring the surface evolution of Centaurs: An empirical test of solar system dynamical history

Observation	Proposal 11995, Observation 7: 2021DB17												Thu Jun 04 15:00:12 GMT 2026
	Diagnostic Status: Warning												
	Observing Template: NIRSpec IFU Spectroscopy												
Diagnostics	(Visit 7:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.												
	(2021DB17 (Obs 7)) Informational (Form): The Visit Planner and Spike may produce different schedulability results.												
Solar System Targets	#	Name	Level 1				Level 2				Level 3		
	(7)	2021DB17	TYPE=ASTEROID,A=19.15690537839405,E=0.3581 399317267978,I=7.508620307976723,O=28.09335608 108688,W=177.3037502569899,M=8.5362258821900 33,EQUINOX=J2000,EPOCH=21-NOV- 2025:00:00:00,EpochTimeScale=TDB										
	Comments: Small Centaur with q>12 AU Extended=NO												
Template	TA Method						HFF Readout Mode						
	NONE						false						
Dithers	#	Dither Type			Size	Starting Point			Number of Points		Points		
	1	4-POINT-DITHER											
Spectral Elements	#	Grating/Filter	Readout Pattern	Groups/Int	Integrations/Exp	Leakcal	Dither	Autocal	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID	
	1	PRISM/CLEAR	NRSIRS2RAPID	58	1	false	true	NONE	4	4	3442.978		
Special Requirements	DEFAULT WINDOW: ANGULAR RATE 2021DB17 FROM JWST LESS THAN 0.075												

Proposal 11995 - Observation 8 - Exploring the surface evolution of Centaurs: An empirical test of solar system dynamical history

Observation	Proposal 11995, Observation 8: 2014UJ225											Thu Jun 04 15:00:12 GMT 2026
	Diagnostic Status: Warning Observing Template: NIRSpec IFU Spectroscopy											
Diagnostics	(Visit 8:1) Warning (Form): Overheads are provisional until the Visit Planner has been run. (2014UJ225 (Obs 8)) Informational (Form): The Visit Planner and Spike may produce different schedulability results.											
Solar System Targets	#	Name	Level 1				Level 2				Level 3	
	(8)	2014UJ225	TYPE=ASTEROID,A=23.42363096764399,E=0.3836 924131393097,I=21.30793360463058,O=18.89883633 455016,W=82.19947636626529,M=1.0498142035173 41,EQUINOX=J2000,EPOCH=21-NOV- 2025:00:00:00,EpochTimeScale=TDB Comments: Small Centaur with q>12 AU Extended=NO									
Template	TA Method						HFF Readout Mode					
	NONE						false					
Dithers	#	Dither Type		Size		Starting Point		Number of Points		Points		
	1	4-POINT-DITHER										
Spectral Elements	#	Grating/Filter	Readout Pattern	Groups/Int	Integrations/Exp	Leakcal	Dither	Autocal	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
	1	PRISM/CLEAR	NRSIRS2RAPID	65	1	false	true	NONE	4	4	3851.467	
Special Requirements	DEFAULT WINDOW: ANGULAR RATE 2014UJ225 FROM JWST LESS THAN 0.075											